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2-4 ONCOTIC PRESSURE

At equilibrium between (1) and (2), we have : $\omega_2 > \omega_1 \Rightarrow H_2O : (1) \rightarrow (2)$

$\Rightarrow$ osmotic pressure: $\Delta\pi = (\omega_2 - \omega_1)RT$

$$\omega_2 - \omega_1 = [(\omega_i)_2 + \omega_p] - (\omega_i)_1$$
$$= [(\omega_i)_2 - (\omega_i)_1] + \omega_p \equiv \Delta\omega_i + \omega_p$$

$$\Rightarrow \Delta\pi = \Delta\omega_i RT + \omega_p RT$$

$\omega_p RT$: ***oncotic*** pressure.

It is the osmotic pressure due to proteins. It expresses the ease with which proteins attract water.

Exercise 13:

A dialysis membrane separates a chamber into two compartments (1) and (2) of equal volume. All experiments are carried out at the same temperature.

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Question 1

In a first experiment, a $NaCl$ solution of concentration C is poured into compartment (1) and another $NaCl$ solution of concentration C' into compartment (2).

At equilibrium, what are the concentrations of sodium and chloride in each compartment?

Responses

$[NaCl]_1 = C$

$[NaCl]_2 = C'$

$NaCl \rightarrow Na^+ + Cl^-$

	(1)	(2)
Initially	$[Na^+]_1 = C$ $[Cl^-]_1 = C$	$[Na^+]_2 = C'$ $[Cl^-]_2 = C'$

$$C_\infty = \frac{C_{1i}+C_{2i}}{2}$$

	(1)	(2)
At equilibrium	$[Na^+]_1 = \frac{C+C'}{2}$ $[Cl^-]_1 = \frac{C+C'}{2}$	$[Na^+]_2 = \frac{C+C'}{2}$ $[Cl^-]_2 = \frac{C+C'}{2}$

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Question 2

In a second experiment, one pours into (1) an undissociated protein of concentration C, and into (2) a $NaCl$ solution of concentration C'.

a) What are the equilibrium concentrations of protein, sodium, and chloride in each compartment ?

b) What pressure must be exerted, and on which compartment, to cancel the water flux?

Responses

a) $[Pr]_1 = C \qquad [NaCl]_2 = C'$

$NaCl \rightarrow Na^+ + Cl^-$

	(1)	(2)
Initially	$[Pr]_1 = C$	$[Na^+]_2 = C'$ $[Cl^-]_2 = C'$

Na^+ and Cl^- diffuse from $(2) \rightarrow (1)$ until equilibrium is reached, i.e., until $[Na^+]_1 = [Na^+]_2$ and $[Cl^-]_1 = [Cl^-]_2$.

$$C_\infty = \frac{C_{1i}+C_{2i}}{2}$$

	(1)	(2)
At equilibrium	$[Na^+]_1 = \frac{C'}{2}$ $[Cl^-]_1 = \frac{C'}{2}$ $[Pr]_1 = C$	$[Na^+]_2 = \frac{C'}{2}$ $[Cl^-]_2 = \frac{C'}{2}$

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b) $\omega_1 > \omega_2 \Rightarrow$ water diffuses from $(2) \rightarrow (1) \Rightarrow \Delta\pi : (2) \rightarrow (1)$

$\Delta\pi = (\Delta\omega_i + \omega_p)RT$

$(\omega_i)_1 = (\omega_{Na^+})_1 + (\omega_{Cl^-})_1 = C' = (\omega_i)_2 \Rightarrow \Delta\omega_i = 0$

$\Rightarrow \Delta\pi = \omega_p RT = CRT$

To prevent the flow of water occurring from $(2) \rightarrow (1)$, it is necessary to apply a pressure equal to the CRT in the direction opposite to $\Delta\pi$, that is, from $(1) \rightarrow (2)$.

Question 3

In a third experiment, one pours into (1) a fully dissociated protein $PrNa_z$, of concentration C, and into (2) a $NaCl$ solution of concentration C'.

a) What are the directions of the initial particle fluxes?

b) What is the ionic concentration of each species at equilibrium? Deduce the osmotic pressure at equilibrium.

c) What is the potential difference between the faces of the membrane? Specify its direction.

Responses

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$a)$ $[NaCl]_2 = C'$

$[PrNa_z]_1 = C$

$PrNa_z \rightarrow Pr^{z-} + zNa^+$

Initially	(1)	(2)
	$[Pr^{z-}]_1 = C$	$[Na^+]_2 = C'$
	$[Na^+]_1 = zC$	$[Cl^-]_2 = C'$

n moles of Cl^- at concentration $x = \frac{n}{V}$ diffuse from $(2) \rightarrow (1)$. They will be accompanied by n moles of Na^+ at the same concentration x.

$b)$

$\Delta\pi = (\Delta\omega_i + \omega_p)RT$

$= [((\omega_i)_1 - (\omega_i)_2) + \omega_p]RT$

$= [(zC + x + x) - (2C' - 2x)) + C]RT$

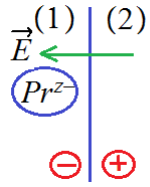
At equilibrium	(1)	(2)
	$[Pr^{z-}]_1 = C$	$[Cl^-]_2 = C' - x$
	$[Cl^-]_1 = x$	$[Na^+]_2 = C' - x$
	$[Na^+]_1 = zC + x$	

$\Rightarrow \Delta\pi = (zC - 2C' + 4x + C)RT$

$c)$ $\vec{E} : (2) \rightarrow (1) \Rightarrow V_2 > V_1$

$\Rightarrow$ The membrane potential is: $V_1 - V_2$

$V_1 - V_2 = -\frac{RT}{zF} \ln \frac{C_1}{C_2}$



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For Cl^-

$$V_1 - V_2 = -\frac{RT}{(-1)F} \ln \frac{x}{C'-x} = -\frac{RT}{F} \ln \frac{C'-x}{x}$$

For Na^+

$$V_1 - V_2 = -\frac{RT}{F} \ln \frac{zC+x}{C'-x}$$

Question 4

In a fourth experiment, a voltage opposite to the previously calculated membrane potential is applied to the system at equilibrium resulting from the third experiment. What are the ionic concentrations in this new equilibrium state?

Responses

Opposite voltage $\Rightarrow V_1 - V_2 = 0 \Rightarrow -\frac{RT}{zF} \ln \frac{C_1}{C_2} = 0 \Rightarrow C_1 = C_2$

$C_\infty = \frac{C_{1i}+C_{2i}}{2} \Rightarrow [Cl^-]_1 = [Cl^-]_2 = \frac{C'-x+x}{2} = \frac{C'}{2}$

$[Na^+]_1 = [Na^+]_2 = \frac{zC+x+C'-x}{2} = \frac{zC+C'}{2}$

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Question 5

In a fifth experiment, one places into (1) a fully dissociated protein $PrNa_z$ at concentration C, and into (2) a solution containing N_aCl at concentration C' as well as the same protein $PrNa_z$ at the same concentration C.

a) What are the equilibrium concentrations ?

b) What is the potential difference across the membrane ?

Responses

a) $[PrNa_z]_1 = C$
$[PrNa_z]_2 = C$
$[NaCl]_2 = C'$

$[Pr^{z-}]_1 = [Pr^{z-}]_2$

$\Rightarrow$ there is no Donnan effect.

$C_\infty = \frac{C_{1i}+C_{2i}}{2}$

	(1)	(2)
Initially	$[Pr^{z-}]_1 = C$ $[Na^+]_1 = zC$	$[Pr^{z-}]_2 = C$ $[Na^+]_2 = C' + zC$ $[Cl^-]_2 = C'$

	(1)	(2)
At equilibrium	$[Pr^{z-}]_1 = C$ $[Na^+]_1 = \frac{C'+2zC}{2}$ $[Cl^-]_1 = \frac{C'}{2}$	$[Pr^{z-}]_2 = C$ $[Na^+]_2 = \frac{C'+2zC}{2}$ $[Cl^-]_2 = \frac{C'}{2}$

b) $C_1 = C_2 \Rightarrow V_1 - V_2 = 0$: the membrane potential is zero.

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Memo

- *For a **semipermeable** membrane*

  The particles that **cannot diffuse** cause water to move and thus generate **osmotic pressure**. They are said to be **osmotically active**.

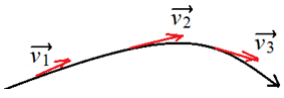
- *For a **dialysis** membrane in **Donnan equilibrium***

  Initially, the diffusible particles diffuse across the membrane until equilibrium is reached.

  At **equilibrium**, the diffusible particles **stop diffusing** and, together with the non-diffusible protein, contribute to the **osmotic pressure**.

2-5 NOTIONS OF FLUID BIOPHYSICS

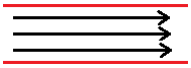
- **Fluids** are liquids and gases.
- **Flow** is the overall motion of a fluid.



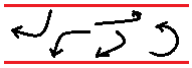
- A **streamline** is a curve that is tangent to the velocity vector at every point.

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- Flow is **laminar** at **low** velocities ⇒ streamlines are **parallel**.



- Flow is **turbulent** at **high** velocities ⇒ **disordered**.



- **Viscosity**: stickiness, **resistance** to the laminar flow of a fluid.

  η : viscosity coefficient of the fluid.

  $[\eta] = Poiseuille\ (Pl).$ $\quad 1Pl = 1Pa.s$

- **Perfect** fluid: a fluid with zero viscosity $\Rightarrow \eta = 0$
- **Real** fluid: a fluid with a non-zero viscosity $\Rightarrow \eta \neq 0$
- **Volumetric flow rate**: $Q_V = \frac{dV}{dt}$ $\quad [Q_v] = m^3/s$

  dV : volume of a fluid passing through a given surface over a period of time dt.